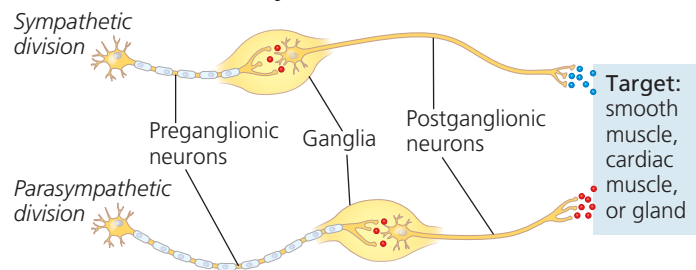


▼ **Figure 49.8 Comparison of pathways in the motor and autonomic nervous systems.**

(a) Motor system



(b) Autonomic nervous system



Key to neurotransmitters • Acetylcholine • Norepinephrine

Glia

The nervous systems of vertebrates and most invertebrates include not only neurons but also **glial cells**, or **glia**. The Schwann cells that myelinate axons in the PNS are an example of glia, as are oligodendrocytes, their counterparts in the CNS. **Figure 49.9** provides an overview of the major types

of glia in the adult vertebrate and the ways in which they nourish, support, and regulate the functioning of neurons.

In embryos, two types of glia play essential roles in the development of the nervous system: radial glia and astrocytes. *Radial glia* form tracks along which newly formed neurons migrate from the neural tube, the structure that gives rise to the CNS (see Figure 47.14). Later, *astrocytes* that are adjacent to brain capillaries participate in the formation of the *blood-brain barrier*, a filtering mechanism that prevents many substances in the blood from entering the CNS. Both radial glia and astrocytes can also act as stem cells, which undergo unlimited cell divisions to self-renew and to form more specialized cells.

CONCEPT CHECK 49.1

1. Which division of the autonomic nervous system would likely be activated if a student learned that an exam she had forgotten about would start in 5 minutes? Explain your answer.
2. **WHAT IF?** Suppose a person had an accident that severed a small nerve required to move some of the fingers of the right hand. Would you also expect an effect on sensation from those fingers?
3. **MAKE CONNECTIONS** Most tissues regulated by the autonomic nervous system receive both sympathetic and parasympathetic input from postganglionic neurons. Responses are typically local. In contrast, the adrenal medulla receives input only from the sympathetic division and only from preganglionic neurons, yet responses are observed throughout the body. Explain why (see Figure 45.19).

For suggested answers, see Appendix A.

▼ **Figure 49.9 Glia in the vertebrate nervous system.**

